

# SHETH MVLU COLLEGE

## SUBJECT:- AI

### Practical 5

AIM:- Feed Forward Backpropagation Neural Network

- Implement the Feed Forward Backpropagation algorithm to train a neural network.
- Use a given dataset to train the neural network for a specific task.
- Evaluate the performance of the trained network on test data.

The screenshot displays a Google Colab notebook interface. The top toolbar includes options for File, Edit, View, Insert, Runtime, Tools, and Help. The left sidebar shows a file explorer with a folder named 'Ai Forward Backpropagation.ipynb'. The main code area contains the following Python code:

```
[5]: # Upload Airbnb dataset in Google Colab
from google.colab import files
uploaded = files.upload()

Choose files | Airbnb_Data.csv
Airbnb_Data.csv(text/csv) - 35913454 bytes, last modified: 03/09/2026 - 100% done
Saving Airbnb_Data.csv to Airbnb_Data.csv

[6]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

[7]: df = pd.read_csv("Airbnb_Data.csv")

print("Dataset Loaded Successfully")
print(df.shape)

Dataset Loaded Successfully
(102599, 26)
/tmp/ipykernel_1372/951088884.py:1: DtypeWarning: Columns (25) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv("Airbnb_Data.csv")
```

The bottom screenshot shows the continuation of the notebook, where the data is preprocessed:

```
[8]: X = df[["price", "minimum nights", "number of reviews",
        "reviews per month", "availability 365"]].copy()
y = df["room type"]

[9]: X["price"] = X["price"].astype(str).str.replace("$", "").str.replace(",", "")
X = X.apply(pd.to_numeric, errors="coerce")

data = pd.concat([X, y], axis=1).dropna()

X = data.drop("room type", axis=1)
y = data["room type"]

le = LabelEncoder()
y = le.fit_transform(y)

print("Data Preprocessing Completed")

Data Preprocessing Completed

[ ] label_encoder = LabelEncoder()
```

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The image displays two screenshots of a Google Colab notebook titled "Ai Forward Backpropagation.ipynb". The notebook is open in a web browser, showing the code editor and the output console.

**First Screenshot:**

```
label_encoder = LabelEncoder()
data[target] = label_encoder.fit_transform(data[target])
print("Room Type Classes:")
for i, class_name in enumerate(label_encoder.classes_):
    print(i, "=", class_name)
```

Output:

```
Room Type Classes:
0 = Entire home/apt
1 = Hotel room
2 = Private room
3 = Shared room
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)
```

Output:

```
Training Data: (68738, 5)
Testing Data: (17185, 5)
```

**Second Screenshot:**

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)
```

Output:

```
Training Data: (68242, 8)
Testing Data: (17061, 8)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
print("Feature Scaling Completed")
```

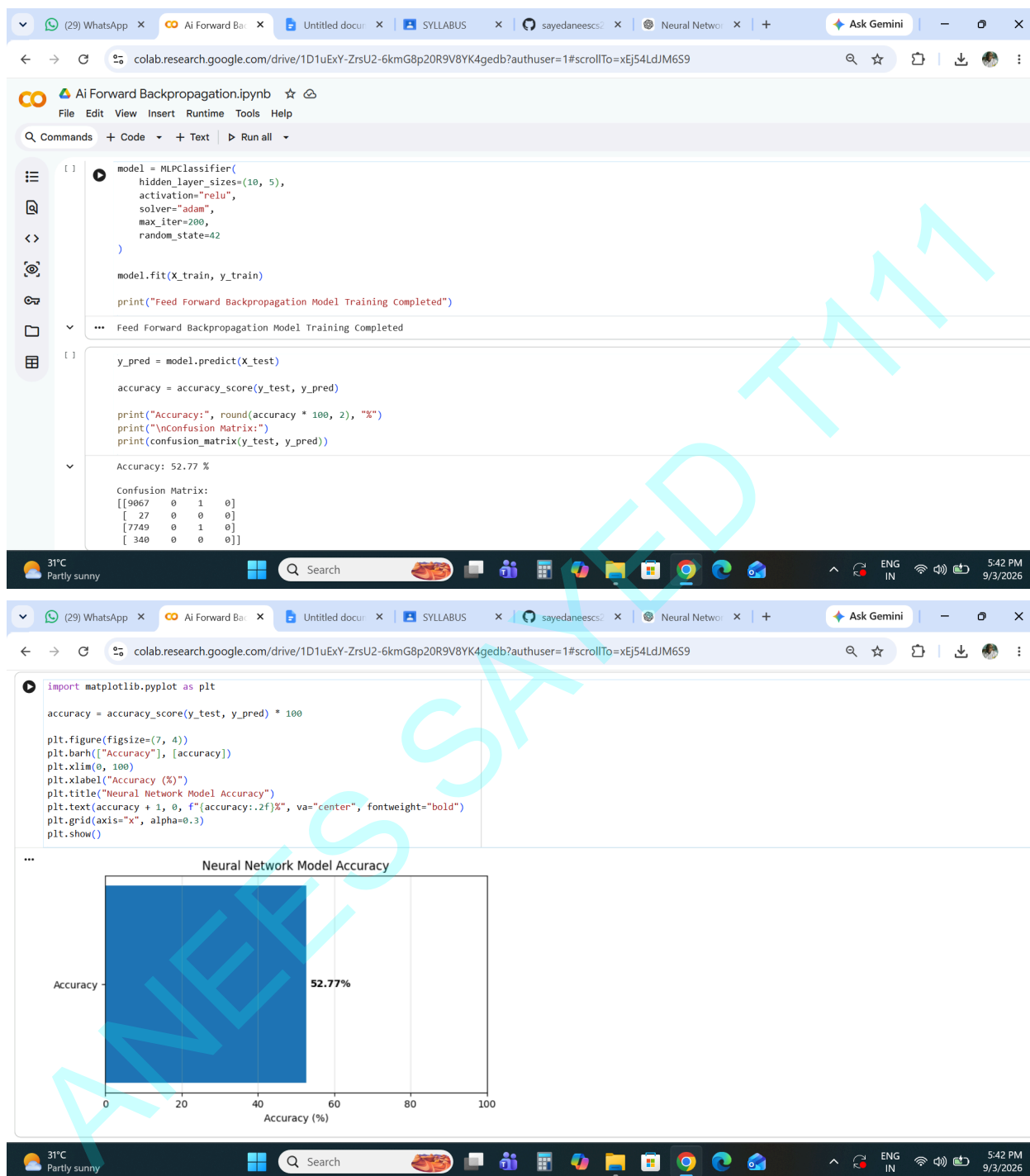
Output:

```
Feature Scaling Completed
```

```
model = MLPClassifier(
```

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